

QT01 – Foundations of Quantum Mechanics

Assessed problem set (PRB) – Week 4

Weight: 30% of module grade

Candidate number: 672128

Question 1 (Canvas question 1)

Question 1

[8 marks]

Consider the operator

$$\hat{A} = \begin{pmatrix} -3 & 2i \\ -2i & -3 \end{pmatrix}.$$

Prove that $\hat{A}$ is Hermitian.

Solution

What is asked. Show that the matrix $\hat{A}$ is equal to its own adjoint.

Principle used. An operator is Hermitian when it coincides with its adjoint,

$$\hat{A} = \hat{A}^\dagger,$$

which is the condition an operator must satisfy in order to represent a physical observable. For an operator acting on a finite-dimensional space, that is, for a matrix, the adjoint is obtained by transposing the matrix and complex conjugating each of its entries,

$$\hat{A}^\dagger = (\hat{A}^\top)^*.$$

The proof therefore consists of carrying out those two operations and comparing the outcome with the original matrix.

Step 1: transpose. Exchanging rows and columns, that is, reflecting the entries about the leading diagonal,

$$\hat{A}^\top = \begin{pmatrix} -3 & 2i \\ -2i & -3 \end{pmatrix}^\top = \begin{pmatrix} -3 & -2i \\ 2i & -3 \end{pmatrix}.$$

Step 2: complex conjugate. Now conjugate every entry, that is, replace i by $-i$ wherever it appears:

$$\hat{A}^\dagger = (\hat{A}^\top)^* = \begin{pmatrix} -3 & -2i \\ 2i & -3 \end{pmatrix}^* = \begin{pmatrix} -3 & 2i \\ -2i & -3 \end{pmatrix}.$$

Step 3: compare. The final matrix is entry by entry identical to $\hat{A}$, so $\hat{A}^\dagger = \hat{A}$ and the operator is Hermitian.

The same statement entry by entry. Writing $\hat{A}^\dagger = \hat{A}$ in components gives the condition $A_{ij} = A_{ji}^*$, which can be checked directly:

$$\begin{array}{lll} A_{11} = -3, & A_{11}^* = -3, & \text{diagonal entry real,} \\ A_{22} = -3, & A_{22}^* = -3, & \text{diagonal entry real,} \\ A_{12} = 2i, & A_{21}^* = (-2i)^* = 2i, & \text{so } A_{12} = A_{21}^*. \end{array}$$

Both diagonal entries are real and the two off-diagonal entries are complex conjugates of one another, which is exactly what Hermiticity demands of a 2×2 matrix.

Check: the eigenvalues must come out real. A Hermitian operator is guaranteed to have real eigenvalues, so computing them is an independent test of the result above. The characteristic equation is

$$\det(\hat{A} - \lambda I) = \begin{vmatrix} -3 - \lambda & 2i \\ -2i & -3 - \lambda \end{vmatrix} = (-3 - \lambda)^2 - (2i)(-2i) = 0.$$

Since $(2i)(-2i) = -4i^2 = 4$, this reduces to

$$(\lambda + 3)^2 = 4 \implies \lambda + 3 = \pm 2,$$

so that

$$\lambda_1 = -1, \quad \lambda_2 = -5.$$

Both are real, as required. A second test is that eigenvectors belonging to distinct eigenvalues of a Hermitian operator must be orthogonal. Solving $(\hat{A} - \lambda I)|\lambda\rangle = 0$ for each eigenvalue gives

$$|\lambda_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix},$$

and their inner product, remembering that forming the bra requires complex conjugation, is

$$\langle \lambda_1 | \lambda_2 \rangle = \frac{1}{2} \begin{pmatrix} -i & 1 \end{pmatrix} \begin{pmatrix} -i \\ 1 \end{pmatrix} = \frac{1}{2} [(-i)(-i) + 1] = \frac{1}{2} (i^2 + 1) = 0.$$

The two eigenvectors are orthogonal, consistent with $\hat{A}$ being Hermitian.

Final answer

$$\hat{A}^\dagger = (\hat{A}^\top)^* = \begin{pmatrix} -3 & 2i \\ -2i & -3 \end{pmatrix} = \hat{A},$$

so $\hat{A}$ is Hermitian. Its eigenvalues, $\lambda_1 = -1$ and $\lambda_2 = -5$, are real and its eigenvectors are orthogonal, as Hermiticity requires.

Physical insight

Hermiticity is the property that makes an operator usable as an observable: measured values are eigenvalues, and a measured value must be a real number. The two structural features seen here, real entries on the diagonal and off-diagonal entries that are complex

conjugate partners, are what enforce this.

It is worth noticing that this particular matrix is built from the spin operators met earlier in the module. Since

$$\hat{A} = -3I - 2\hat{\sigma}_y, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

and $\hat{\sigma}_y$ has eigenvalues ± 1 , the eigenvalues of $\hat{A}$ must be -3 ∓ 2 , that is -5 and -1 . This reproduces the characteristic-equation result without any determinant, and shows that $\hat{A}$ describes a spin observable that has been shifted by a constant -3 and rescaled by a factor -2 . Adding a multiple of the identity shifts every eigenvalue but changes no eigenvector, which is why the physics of such an operator is the physics of $\hat{\sigma}_y$ in disguise.

Question 2 (Canvas questions 2–5)

Question 2

[37 marks]

The wave function of a particle moving in one dimension is given by

$$\psi(x) = \begin{cases} B \exp(\beta x) & \text{for } x < 0, \\ B \exp(-2\beta x) & \text{for } x \geq 0, \end{cases}$$

where β is a real and positive constant.

- What is the probability density for finding the particle along the x -axis? Sketch the probability density function. [7 marks]
- Calculate the normalisation constant B . [10 marks]
- Calculate the average position of the particle on the x -axis, as a function of β . Mark the average position in your sketch. [15 marks]
- Mark the most likely position of the particle in your sketch. [5 marks]

Hint: You can calculate the integrals you need by expressing powers of x through repeated differentiation with respect to the parameter in the exponential, for example

$$\begin{aligned} \int_a^b x \exp(\gamma x) \, dx &= \frac{\partial}{\partial \gamma} \int_a^b \exp(\gamma x) \, dx, \\ \int_a^b x^2 \exp(\gamma x) \, dx &= \frac{\partial^2}{\partial \gamma^2} \int_a^b \exp(\gamma x) \, dx, \end{aligned}$$

and so on.

Solution

(a) The probability density, and its sketch. (Canvas Q2)

Principle used. The wave function itself is not directly measurable. What is measurable is the probability density for finding the particle at position x , given by the modulus square of the wave function,

$$\rho(x) = |\psi(x)|^2 = \psi^*(x)\psi(x),$$

so that $\rho(x) \, dx$ is the probability of finding the particle in the interval between x and $x + dx$. This is the position-space statement of the Born rule.

Applying it. The exponentials are real for real x and β , so squaring the modulus only affects the constant B , giving $|B|^2$. Squaring each branch separately,

$$\rho(x) = |\psi(x)|^2 = \begin{cases} |B|^2 \exp(2\beta x), & x < 0, \\ |B|^2 \exp(-4\beta x), & x \geq 0. \end{cases}$$

Note that each exponent has doubled, because $|e^{\alpha x}|^2 = e^{2\alpha x}$ for real α .

Features that the sketch must show. Three properties fix the shape of the curve.

- It is continuous at the origin and peaks there.* Approaching from either side, $\rho(0^-) = \rho(0^+) = |B|^2$, so the two branches join. Since $\beta > 0$, the left branch

$e^{2\beta x}$ rises as x increases towards 0, and the right branch $e^{-4\beta x}$ falls as x increases away from 0. The density therefore has its largest value at $x = 0$.

- (b) *It decays to zero in both directions*, as $e^{2\beta x} \rightarrow 0$ when $x \rightarrow -\infty$ and $e^{-4\beta x} \rightarrow 0$ when $x \rightarrow +\infty$. This is what makes the function normalisable, and it is checked explicitly in part (b).
- (c) *It is asymmetric*. The characteristic decay length of the density is $1/(2\beta)$ on the left but only $1/(4\beta)$ on the right, these being the distances over which ρ falls by a factor of e . The density is therefore spread twice as far to the left of the origin as to the right, and the curve is visibly lopsided.

A fourth feature is worth recording because it is easy to miss. The slopes of the two branches at the origin are

$$\left. \frac{d\rho}{dx} \right|_{0-} = 2\beta|B|^2, \quad \left. \frac{d\rho}{dx} \right|_{0+} = -4\beta|B|^2,$$

which are not equal. The maximum at $x = 0$ is therefore a sharp corner, or cusp, rather than a smooth turning point. This matters in part (d).

The resulting curve is drawn in Fig. (1), where the normalisation constant found in part (b) has been used to fix the vertical scale, and where the average and most likely positions obtained in parts (c) and (d) are marked.

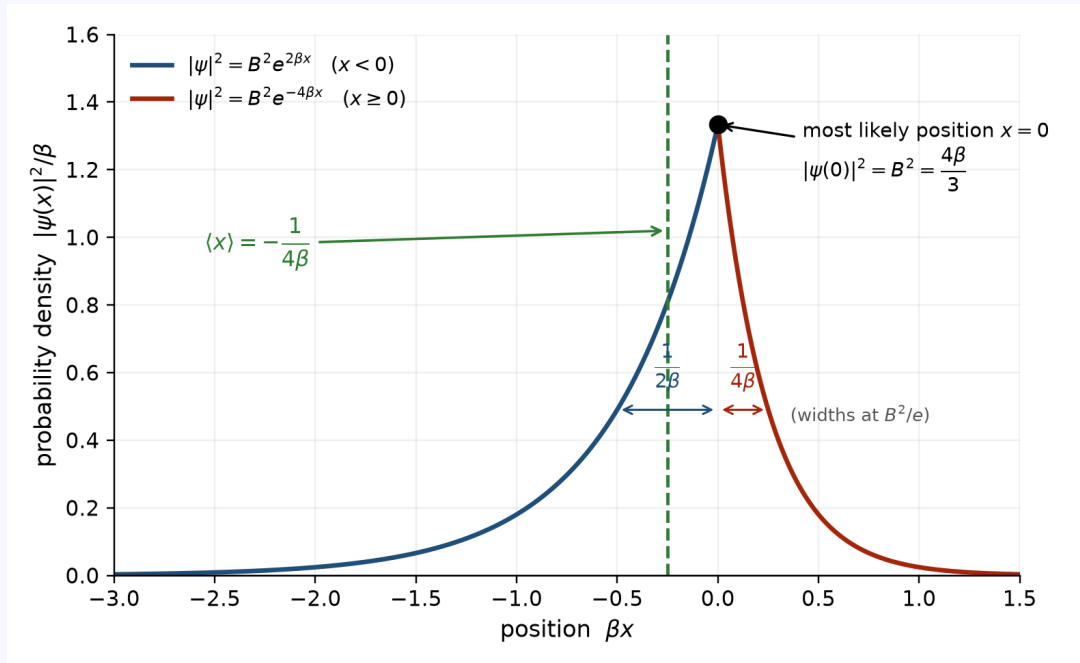


Figure 1: The probability density $|\psi(x)|^2$, plotted in dimensionless units so that the curve is valid for any $\beta > 0$: the horizontal axis is βx and the vertical axis is $|\psi(x)|^2/\beta$. The density rises as $e^{2\beta x}$ for $x < 0$, falls as $e^{-4\beta x}$ for $x \geq 0$, and peaks at the cusp at $x = 0$ with the value $B^2 = 4\beta/3$. The horizontal arrows mark the decay lengths $1/(2\beta)$ and $1/(4\beta)$, at which the density has fallen to B^2/e ; the left-hand tail is twice as extended as the right-hand one. The dashed vertical line marks the average position $\langle x \rangle = -1/(4\beta)$, which lies to the left of the peak because of that asymmetry.

(b) The normalisation constant B .**(Canvas Q3)**

Principle used. The particle must be found somewhere on the x -axis, so the total probability must equal one:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1.$$

This condition determines $|B|$.

Splitting the integral. The wave function is defined by different expressions on either side of the origin, so the integral splits at $x = 0$:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = |B|^2 \int_{-\infty}^0 e^{2\beta x} dx + |B|^2 \int_0^{\infty} e^{-4\beta x} dx.$$

The two integrals. Both are elementary. For the left-hand one, the exponential vanishes at the lower limit because $\beta > 0$:

$$\int_{-\infty}^0 e^{2\beta x} dx = \left[\frac{e^{2\beta x}}{2\beta} \right]_{-\infty}^0 = \frac{1}{2\beta} - 0 = \frac{1}{2\beta}.$$

For the right-hand one, the exponential vanishes at the upper limit:

$$\int_0^{\infty} e^{-4\beta x} dx = \left[-\frac{e^{-4\beta x}}{4\beta} \right]_0^{\infty} = 0 - \left(-\frac{1}{4\beta} \right) = \frac{1}{4\beta}.$$

The right-hand contribution is half the left-hand one, which is the quantitative form of the asymmetry noted in part (a).

Solving for B . Adding the two contributions and imposing normalisation,

$$|B|^2 \left(\frac{1}{2\beta} + \frac{1}{4\beta} \right) = |B|^2 \frac{2+1}{4\beta} = \frac{3|B|^2}{4\beta} = 1,$$

so that

$$|B|^2 = \frac{4\beta}{3}, \quad |B| = \sqrt{\frac{4\beta}{3}} = 2\sqrt{\frac{\beta}{3}}.$$

Only the modulus of B is fixed by this condition: multiplying ψ by a phase factor $e^{i\theta}$ leaves $|\psi|^2$, and hence every measurable prediction, unchanged. It is conventional to take that free phase to be zero and to quote the real, positive root,

$$B = 2\sqrt{\frac{\beta}{3}} = \frac{2\sqrt{\beta}}{\sqrt{3}}.$$

Check: dimensions. The quantity β appears in the combination βx , which must be dimensionless, so β has dimensions of inverse length. Then $|B| = \sqrt{4\beta/3}$ has dimensions of $(\text{length})^{-1/2}$, which is exactly right for a one-dimensional wave function, since $|\psi|^2 dx$ must be a pure number. Using this value, the peak of the density is $\rho(0) = |B|^2 = 4\beta/3$, which is the height marked in Fig. (1).

(c) The average position.**(Canvas Q4)**

Principle used. The average, or expectation, value of an observable in a normalised state is obtained by sandwiching the corresponding operator between the state and its

conjugate. For position, whose operator acts simply by multiplication, $\hat{x}\psi(x) = x\psi(x)$, this reads

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{x} \psi(x) dx = \int_{-\infty}^{\infty} x |\psi(x)|^2 dx.$$

This is the familiar mean of a distribution, with $|\psi(x)|^2$ playing the role of the probability density. It is legitimate to use this form, rather than dividing by $\langle \psi | \psi \rangle$, precisely because B has already been chosen in part (b) to make the state normalised.

Setting up. Splitting at the origin as before,

$$\langle \hat{x} \rangle = |B|^2 \underbrace{\int_{-\infty}^0 x e^{2\beta x} dx}_{I_-} + |B|^2 \underbrace{\int_0^{\infty} x e^{-4\beta x} dx}_{I_+}.$$

Evaluating the integrals by differentiating with respect to the exponent.

Following the hint, a factor of x in the integrand can be generated by differentiating the corresponding exponential integral with respect to the parameter γ in $e^{\gamma x}$, because $\partial e^{\gamma x} / \partial \gamma = x e^{\gamma x}$.

For the left-hand integral, define, for $\gamma > 0$ so that the integral converges,

$$F(\gamma) \equiv \int_{-\infty}^0 e^{\gamma x} dx = \left[\frac{e^{\gamma x}}{\gamma} \right]_{-\infty}^0 = \frac{1}{\gamma}.$$

Differentiating both sides with respect to γ ,

$$\int_{-\infty}^0 x e^{\gamma x} dx = \frac{\partial}{\partial \gamma} \left(\frac{1}{\gamma} \right) = -\frac{1}{\gamma^2},$$

and setting $\gamma = 2\beta$ gives

$$I_- = -\frac{1}{(2\beta)^2} = -\frac{1}{4\beta^2}.$$

For the right-hand integral, define, for $\gamma < 0$ so that the integral converges,

$$G(\gamma) \equiv \int_0^{\infty} e^{\gamma x} dx = \left[\frac{e^{\gamma x}}{\gamma} \right]_0^{\infty} = -\frac{1}{\gamma}.$$

Differentiating with respect to γ ,

$$\int_0^{\infty} x e^{\gamma x} dx = \frac{\partial}{\partial \gamma} \left(-\frac{1}{\gamma} \right) = \frac{1}{\gamma^2},$$

and setting $\gamma = -4\beta$ gives

$$I_+ = \frac{1}{(-4\beta)^2} = \frac{1}{16\beta^2}.$$

Combining. The two contributions have opposite signs, since the particle can be found on either side of the origin, and they do not cancel:

$$I_- + I_+ = -\frac{1}{4\beta^2} + \frac{1}{16\beta^2} = \frac{-4 + 1}{16\beta^2} = -\frac{3}{16\beta^2}.$$

Inserting $|B|^2 = 4\beta/3$ from part (b),

$$\langle \hat{x} \rangle = \frac{4\beta}{3} \times \left(-\frac{3}{16\beta^2} \right) = -\frac{12\beta}{48\beta^2} = -\frac{1}{4\beta}.$$

Check 1: the same integrals, done by parts. Evaluating I_+ independently, by parts with $u = x$ and $dv = e^{-4\beta x} dx$,

$$I_+ = \left[-\frac{x e^{-4\beta x}}{4\beta} \right]_0^\infty + \frac{1}{4\beta} \int_0^\infty e^{-4\beta x} dx = 0 + \frac{1}{4\beta} \cdot \frac{1}{4\beta} = \frac{1}{16\beta^2},$$

the boundary term vanishing at both limits because the exponential decays faster than x grows. Substituting $x = -u$ in I_- turns it into $-\int_0^\infty u e^{-2\beta u} du = -1/(4\beta^2)$ by the same standard result. Both agree with the values found above.

Check 2: sign and size. The answer is negative, which is what the sketch demands: the density is spread twice as far to the left of the origin as to the right, so the mean must be pulled to the negative side of the peak. Its magnitude, $1/(4\beta)$, is smaller than the left-hand decay length $1/(2\beta)$, as it must be, since the mean cannot lie beyond the bulk of the distribution. Finally, $1/\beta$ has dimensions of length, so the result is dimensionally a position.

The value $\langle x \rangle = -1/(4\beta)$, that is $\beta x = -0.25$ in the dimensionless units of the figure, is marked by the dashed vertical line in Fig. (1).

(d) The most likely position.

(Canvas Q5)

Principle used. The most likely position is the value of x at which the probability density is largest, that is, the mode of the distribution $\rho(x) = |\psi(x)|^2$.

Locating the maximum. The usual route of solving $d\rho/dx = 0$ fails here, and it is worth being explicit about why. Differentiating each branch,

$$\frac{d\rho}{dx} = \begin{cases} 2\beta|B|^2 e^{2\beta x} > 0, & x < 0, \\ -4\beta|B|^2 e^{-4\beta x} < 0, & x > 0, \end{cases}$$

and neither expression is ever zero, because an exponential never vanishes at finite argument and $\beta > 0$. There is therefore no stationary point anywhere.

The maximum is instead found by monotonicity. The density increases strictly on the whole of $x < 0$ and decreases strictly on the whole of $x > 0$, so its largest value must occur where the two branches meet, at

$$x_{\max} = 0,$$

where the density takes the value

$$\rho(0) = |B|^2 = \frac{4\beta}{3}.$$

Because the left-hand slope $2\beta|B|^2$ and the right-hand slope $-4\beta|B|^2$ differ, the curve turns over in a sharp corner rather than a smooth arch. This is the cusp visible in Fig. (1), where the most likely position is marked with a filled circle at the apex.

Final answer

(a) The probability density is

$$|\psi(x)|^2 = \begin{cases} |B|^2 e^{2\beta x}, & x < 0, \\ |B|^2 e^{-4\beta x}, & x \geq 0, \end{cases}$$

a cusped, asymmetric peak centred on the origin, sketched in Fig. (1).

(b) $|B|^2 = \frac{4\beta}{3}$, so $B = 2\sqrt{\frac{\beta}{3}}$ taking the phase to be zero.

(c) $\langle x \rangle = -\frac{1}{4\beta}$, marked by the dashed line in Fig. (1).

(d) The most likely position is $x = 0$, where $|\psi(0)|^2 = 4\beta/3$ is greatest; it is marked by the filled circle in Fig. (1).

Physical insight

The striking feature of this state is that the average position and the most likely position are different: $\langle x \rangle = -1/(4\beta)$ while the mode sits at $x = 0$. There is no contradiction. The mean answers “where does the particle land on average over many measurements”, and it is sensitive to the long left-hand tail, which contributes many detections far out at negative x . The mode answers “which single location is most likely in any one measurement”, and it ignores the tails entirely. The two coincide only for a symmetric distribution, and this one is deliberately lopsided, the density falling off twice as fast to the right as to the left.

A useful consequence is that neither number on its own describes the state. The spread does the rest of the work. Using the same parameter-differentiation trick a second time gives $\langle x^2 \rangle = 3/(8\beta^2)$, and hence

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{3}{8\beta^2} - \frac{1}{16\beta^2}} = \frac{\sqrt{5}}{4\beta} \approx \frac{0.56}{\beta},$$

which is comfortably larger than $|\langle x \rangle|$. The particle is genuinely delocalised over a region of order $1/\beta$, and quoting $\langle x \rangle$ alone would suggest a far more definite position than the state actually possesses.

Finally, the cusp carries physical content. A kink in ψ , meaning a discontinuous first derivative, is not allowed for a particle moving in a smooth potential; it arises only where the potential is infinitely sharp. A wave function of exactly this two-sided exponential form is what a delta-function potential well at the origin produces, and the fact that the two decay constants differ signals that the potential is not symmetric about that point.

Question 3 (Canvas questions 6–9)

Question 3

[22 marks]

Take the wave function

$$\varphi(x) = N \sin(kx).$$

- (a) Calculate the effect of the operator

$$\left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right]$$

on $\varphi(x)$.

[2 marks]

- (b) Calculate the effect of the operator

$$\left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right]$$

on $\varphi(x)$, that is, multiply $\varphi(x)$ by x before acting with the differential operator.

[5 marks]

- (c) Calculate the difference of the two previous results, that is,

result of (a) – result of (b).

Express the answer in terms of $\varphi(x)$.

[5 marks]

- (d) Repeat parts (a) to (c) for a general wave function $\psi(x)$, that is, calculate

$$\left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right] \psi(x) - \left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right] \psi(x).$$

[10 marks]

Solution

Notation. The differential operator appearing throughout is the momentum operator,

$$\hat{p} = \frac{\hbar}{i} \frac{\partial}{\partial x} = -i\hbar \frac{\partial}{\partial x},$$

using $1/i = -i$. The other operator is the position operator $\hat{x}$, which acts by multiplication, $\hat{x}\psi(x) = x\psi(x)$. Part (a) therefore asks for $\hat{x}\hat{p}\varphi$ and part (b) for $\hat{p}\hat{x}\varphi$; the two differ only in which operator acts first, and the point of the question is that this order matters.

(a) The operator $x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right)$ acting on φ . (Canvas Q6)

Here the derivative acts on φ first, and the result is then multiplied by x . Differentiating,

$$\frac{\partial \varphi}{\partial x} = \frac{\partial}{\partial x} (N \sin(kx)) = Nk \cos(kx),$$

so that

$$\left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right] \varphi(x) = x \cdot \frac{\hbar}{i} Nk \cos(kx) = -i\hbar Nkx \cos(kx).$$

(b) The operator $\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x$ acting on φ . **(Canvas Q7)**

Now the multiplication by x happens first, so the derivative acts on the product $x\varphi(x)$ and the product rule is needed:

$$\frac{\partial}{\partial x} \left(x \varphi(x) \right) = \varphi(x) + x \frac{\partial \varphi}{\partial x} = N \sin(kx) + Nkx \cos(kx).$$

Multiplying by the prefactor,

$$\left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right] \varphi(x) = \frac{\hbar}{i} \left(N \sin(kx) + Nkx \cos(kx) \right) = -i\hbar N \sin(kx) - i\hbar Nkx \cos(kx).$$

The second term reproduces the answer to part (a); the first term is entirely new, and it is the term generated by the product rule.

(c) The difference of the two results. **(Canvas Q8)**

Subtracting the result of (b) from the result of (a), the $x \cos(kx)$ terms cancel because they are identical:

$$\begin{aligned} (a) - (b) &= -i\hbar Nkx \cos(kx) - \left(-i\hbar N \sin(kx) - i\hbar Nkx \cos(kx) \right) \\ &= -i\hbar Nkx \cos(kx) + i\hbar N \sin(kx) + i\hbar Nkx \cos(kx) \\ &= i\hbar N \sin(kx). \end{aligned}$$

The surviving term is proportional to the original wave function, since $\varphi(x) = N \sin(kx)$:

$$(a) - (b) = i\hbar \varphi(x).$$

The difference has returned the wave function unchanged apart from a constant factor $i\hbar$. Note in particular that the answer does not depend on k or on N .

(d) The same calculation for a general wave function. **(Canvas Q9)**

Why repeat it. The result of part (c) was obtained for one particular choice of φ . If it holds for an arbitrary wave function, it is a statement about the operators themselves rather than about the state they act on, which is a far stronger conclusion.

The calculation. Let $\psi(x)$ be any differentiable wave function. The first term is the derivative of ψ , multiplied by x :

$$\left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right] \psi(x) = \frac{\hbar}{i} x \frac{\partial \psi}{\partial x}.$$

The second term requires the product rule, exactly as in part (b):

$$\left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right] \psi(x) = \frac{\hbar}{i} \frac{\partial}{\partial x} \left(x \psi(x) \right) = \frac{\hbar}{i} \left(\psi(x) + x \frac{\partial \psi}{\partial x} \right).$$

Subtracting, the terms containing $x \partial\psi/\partial x$ cancel:

$$\begin{aligned} \left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right] \psi(x) - \left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right] \psi(x) &= \frac{\hbar}{i} \left(x \frac{\partial\psi}{\partial x} - \psi(x) - x \frac{\partial\psi}{\partial x} \right) \\ &= -\frac{\hbar}{i} \psi(x) \\ &= i\hbar \psi(x), \end{aligned}$$

where the last line uses $-1/i = i$, or equivalently $\hbar/i = -i\hbar$ so that $-\hbar/i = +i\hbar$.

What this means. Since ψ was arbitrary, the two operators can be compared directly, without reference to any state. Writing the difference as a commutator,

$$[\hat{x}, \hat{p}] \psi(x) = (\hat{x}\hat{p} - \hat{p}\hat{x}) \psi(x) = i\hbar \psi(x) \quad \implies \quad [\hat{x}, \hat{p}] = i\hbar.$$

This is the canonical commutation relation between position and momentum.

Check: consistency with part (c). Setting $\psi(x) = \varphi(x) = N \sin(kx)$ in the general result gives $i\hbar\varphi(x)$, which is exactly what the explicit calculation in part (c) produced. The special case and the general case agree.

Check: dimensions. The commutator of a position and a momentum has dimensions of length $\times$ momentum, which is an action, and $\hbar$ is indeed an action. The result is dimensionally consistent.

Final answer

(a) $\left[x \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \right] \varphi(x) = -i\hbar N k x \cos(kx).$

(b) $\left[\left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) x \right] \varphi(x) = -i\hbar N \sin(kx) - i\hbar N k x \cos(kx).$

(c) (a) $-$ (b) $= i\hbar N \sin(kx) = i\hbar \varphi(x).$

(d) For an arbitrary $\psi(x)$ the same difference gives $i\hbar \psi(x)$, that is, the operator identity $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar.$

Physical insight

The whole result comes from one extra term in the product rule, and yet it is the point at which quantum mechanics parts company with classical mechanics. Classically, position and momentum are numbers, and numbers commute: $xp - px = 0$ identically. Here the same combination is not zero but $i\hbar$, a constant so small on everyday scales that the classical answer is recovered for all practical purposes, and so non-zero that the microscopic world is reorganised around it.

Two consequences follow. First, order of operations now carries physical meaning. Measuring position and then momentum is not the same operation as measuring momentum and then position, and the difference between the two is precisely $i\hbar$. Second, and equivalently, $\hat{x}$ and $\hat{p}$ can have no common eigenstates. If some state ψ were an eigenstate of both, with $\hat{x}\psi = x_0\psi$ and $\hat{p}\psi = p_0\psi$, then $(\hat{x}\hat{p} - \hat{p}\hat{x})\psi = (x_0p_0 - p_0x_0)\psi = 0$, contradicting $i\hbar\psi \neq 0$. No state can therefore have a sharp position and a sharp momentum at the same time.

Quantitatively, the general uncertainty relation $\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$ turns this com-

mutator directly into the Heisenberg relation

$$\Delta x \Delta p \geq \frac{\hbar}{2},$$

so the small algebraic remainder computed in part (d) is the origin of the limit on how sharply a particle can be localised. The state of Question 2 illustrates the same point from the other side: it has a well-defined spread $\Delta x = \sqrt{5}/(4\beta)$ in position, and the price of that localisation is an unavoidable spread in momentum.

Question 4 (Canvas question 10)

Question 4

[8 marks]

A system is prepared in a state

$$|\psi\rangle = \frac{1}{\sqrt{3}} |\phi_1\rangle - \frac{\sqrt{2}}{\sqrt{3}} |\phi_2\rangle + \frac{i}{\sqrt{3}} |\phi_3\rangle,$$

where $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ are orthonormal eigenstates of an observable A , with corresponding eigenvalues a_1 , a_2 , and a_3 , respectively. What is the probability of measuring A to be a_3 ?

Solution

What is asked. Find the probability that a measurement of the observable A on the state $|\psi\rangle$ returns the eigenvalue a_3 .

Principle used. For a *normalised* state expanded in the orthonormal eigenstates of the observable,

$$|\psi\rangle = \sum_n \alpha_n |\phi_n\rangle, \quad \hat{A} |\phi_n\rangle = a_n |\phi_n\rangle,$$

the Born rule gives the probability of the outcome a_n as

$$P(a_n) = |\langle \phi_n | \psi \rangle|^2 = |\alpha_n|^2.$$

The requirement that the state be normalised is essential, and it is the first thing to check.

Step 1: is the state normalised? Reading off the expansion coefficients,

$$\alpha_1 = \frac{1}{\sqrt{3}}, \quad \alpha_2 = -\frac{\sqrt{2}}{\sqrt{3}}, \quad \alpha_3 = \frac{i}{\sqrt{3}},$$

and using orthonormality, $\langle \phi_m | \phi_n \rangle = \delta_{mn}$, so that all cross terms vanish,

$$\langle \psi | \psi \rangle = \sum_n |\alpha_n|^2 = \left| \frac{1}{\sqrt{3}} \right|^2 + \left| -\frac{\sqrt{2}}{\sqrt{3}} \right|^2 + \left| \frac{i}{\sqrt{3}} \right|^2 = \frac{1}{3} + \frac{2}{3} + \frac{1}{3} = \frac{4}{3} \neq 1.$$

Note that the modulus square removes both the minus sign of α_2 and the factor i in α_3 , since $|-z|^2 = |z|^2$ and $|i|^2 = i i^* = 1$. The state as given is therefore *not* normalised, and the Born rule cannot be applied to it directly.

Step 2: account for the normalisation. There are two equivalent ways to proceed. *Method 1: divide by the norm.* For an un-normalised state the probability is the squared overlap divided by $\langle \psi | \psi \rangle$,

$$P(a_3) = \frac{|\langle \phi_3 | \psi \rangle|^2}{\langle \psi | \psi \rangle}.$$

The overlap is extracted using orthonormality, only the $n = 3$ term surviving:

$$\langle \phi_3 | \psi \rangle = \frac{1}{\sqrt{3}} \langle \phi_3 | \phi_1 \rangle - \frac{\sqrt{2}}{\sqrt{3}} \langle \phi_3 | \phi_2 \rangle + \frac{i}{\sqrt{3}} \langle \phi_3 | \phi_3 \rangle = \frac{i}{\sqrt{3}},$$

so that $|\langle\phi_3|\psi\rangle|^2 = 1/3$ and

$$P(a_3) = \frac{1/3}{4/3} = \frac{1}{4}.$$

Method 2: normalise the state first. Dividing by $\sqrt{\langle\psi|\psi\rangle} = 2/\sqrt{3}$ gives the normalised state

$$|\psi_{\text{norm}}\rangle = \frac{\sqrt{3}}{2} |\psi\rangle = \frac{1}{2} |\phi_1\rangle - \frac{\sqrt{2}}{2} |\phi_2\rangle + \frac{i}{2} |\phi_3\rangle,$$

whose coefficients now satisfy $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. Applying the Born rule to this state,

$$P(a_3) = \left|\frac{i}{2}\right|^2 = \frac{1}{4},$$

in agreement with Method 1.

Check: the probabilities must sum to one. Computing the other two outcomes the same way,

$$P(a_1) = \frac{1/3}{4/3} = \frac{1}{4}, \quad P(a_2) = \frac{2/3}{4/3} = \frac{1}{2}, \quad P(a_3) = \frac{1}{4},$$

and indeed

$$P(a_1) + P(a_2) + P(a_3) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

Each probability also lies in the interval $[0, 1]$, as any probability must. Had the normalisation step been skipped, the three squared coefficients would have summed to $4/3$, which is impossible for a set of exclusive probabilities, and that alone would have flagged the error.

Final answer

The state as given is not normalised, since $\langle\psi|\psi\rangle = 4/3$. Allowing for this,

$$P(a_3) = \frac{|\langle\phi_3|\psi\rangle|^2}{\langle\psi|\psi\rangle} = \frac{1/3}{4/3} = \frac{1}{4}.$$

The probability of measuring A to be a_3 is $1/4$, that is, 25%. For completeness, $P(a_1) = 1/4$ and $P(a_2) = 1/2$, and the three sum to one.

Physical insight

Two features of this calculation are worth drawing out.

First, the factor i multiplying $|\phi_3\rangle$ has no effect whatever on the answer, because the modulus square discards it. This is an instance of a general rule: the probabilities for measuring A depend only on the magnitudes of the expansion coefficients in the eigenbasis of $\hat{A}$. It would be wrong, however, to conclude that the phase is meaningless. Relative phases between components are physically real and show up as interference as soon as a different observable is measured, one whose eigenstates are superpositions of the $|\phi_n\rangle$. The same is true of the minus sign on $|\phi_2\rangle$. A phase is invisible to one measurement and decisive for another.

Second, normalisation is not a cosmetic step. The squared coefficients are probabilities

only when the state has unit length, because the total probability of all mutually exclusive outcomes must be one. Here the raw coefficients would have given $1/3$ for the third outcome and a total of $4/3$ across all three, which is not a probability distribution at all. Dividing by $\langle\psi|\psi\rangle$ restores the interpretation, and the final check that the probabilities sum to one is the cheapest way to confirm that it has been done correctly.

Question 5 (Canvas questions 11–13)

Question 5

[25 marks]

A system is initially in the state

$$|\psi_0\rangle = \frac{1}{\sqrt{7}} \left[\sqrt{2} |\phi_1\rangle + \sqrt{3} |\phi_2\rangle + |\phi_3\rangle + |\phi_4\rangle \right],$$

where $|\phi_n\rangle$ are eigenstates of the system's Hamiltonian such that

$$\hat{H} |\phi_n\rangle = n^2 \mathcal{E}_0 |\phi_n\rangle.$$

- (a) If energy is measured, what values will be obtained and with what probabilities? [10 marks]

- (b) Consider an operator $\hat{A}$ whose action on $|\phi_n\rangle$ is defined by

$$\hat{A} |\phi_n\rangle = (n+1)a_0 |\phi_n\rangle.$$

If A is measured, what values will be obtained and with what probabilities? [10 marks]

- (c) Suppose that a measurement of the energy yields $4\mathcal{E}_0$. If we measure A immediately afterwards, what value will be obtained? [5 marks]

Solution

Preliminary: is the state normalised? Every part of this question uses the Born rule, which requires a normalised state, so this is checked once at the outset. The expansion coefficients are

$$\alpha_1 = \frac{\sqrt{2}}{\sqrt{7}}, \quad \alpha_2 = \frac{\sqrt{3}}{\sqrt{7}}, \quad \alpha_3 = \frac{1}{\sqrt{7}}, \quad \alpha_4 = \frac{1}{\sqrt{7}},$$

and, since the $|\phi_n\rangle$ are orthonormal so that all cross terms vanish,

$$\langle\psi_0|\psi_0\rangle = \sum_{n=1}^4 |\alpha_n|^2 = \frac{2}{7} + \frac{3}{7} + \frac{1}{7} + \frac{1}{7} = \frac{7}{7} = 1.$$

The state is already normalised, so the Born rule may be applied directly, with no rescaling.

(a) Measuring the energy. (Canvas Q11)

Principle used. Two postulates are needed. The first states that the only possible outcomes of measuring an observable are the eigenvalues of the corresponding operator; for the energy, that operator is the Hamiltonian $\hat{H}$. The second, the Born rule, gives the probability of each outcome as the squared modulus of the corresponding expansion coefficient, $P(E_n) = |\langle\phi_n|\psi_0\rangle|^2 = |\alpha_n|^2$.

The possible outcomes. The eigenvalue equation is given as $\hat{H} |\phi_n\rangle = n^2 \mathcal{E}_0 |\phi_n\rangle$, so the energy eigenvalues are $E_n = n^2 \mathcal{E}_0$. Only the four states appearing in $|\psi_0\rangle$ can be

obtained, since the coefficients of all other eigenstates are zero. For $n = 1, 2, 3, 4$,

$$E_1 = \mathcal{E}_0, \quad E_2 = 4\mathcal{E}_0, \quad E_3 = 9\mathcal{E}_0, \quad E_4 = 16\mathcal{E}_0.$$

The probabilities. Taking the squared modulus of each coefficient,

$$P(\mathcal{E}_0) = \frac{2}{7}, \quad P(4\mathcal{E}_0) = \frac{3}{7}, \quad P(9\mathcal{E}_0) = \frac{1}{7}, \quad P(16\mathcal{E}_0) = \frac{1}{7}.$$

Collecting the results:

Energy E_n	$\mathcal{E}_0$	$4\mathcal{E}_0$	$9\mathcal{E}_0$	$16\mathcal{E}_0$
Probability $P(E_n)$	$2/7$	$3/7$	$1/7$	$1/7$

Check. The probabilities sum to $2/7 + 3/7 + 1/7 + 1/7 = 1$, and each lies between 0 and 1. As a further test, the expectation value

$$\langle \hat{H} \rangle = \sum_n P(E_n) E_n = \frac{2(1) + 3(4) + 1(9) + 1(16)}{7} \mathcal{E}_0 = \frac{39}{7} \mathcal{E}_0 \approx 5.6 \mathcal{E}_0$$

lies between the smallest and largest possible outcomes, $\mathcal{E}_0$ and $16\mathcal{E}_0$, as an average must.

(b) Measuring A .

(Canvas Q12)

The key observation. The operator $\hat{A}$ is defined by $\hat{A}|\phi_n\rangle = (n+1)a_0|\phi_n\rangle$, which says that the states $|\phi_n\rangle$ are eigenstates of $\hat{A}$ as well as of $\hat{H}$. The state $|\psi_0\rangle$ is therefore *already* expanded in the eigenbasis of $\hat{A}$, and no change of basis is needed: the expansion coefficients, and hence the probabilities, are exactly those used in part (a). Only the list of eigenvalues changes.

The possible outcomes. The eigenvalues of $\hat{A}$ are $a_n = (n+1)a_0$, so for $n = 1, 2, 3, 4$,

$$a_1 = 2a_0, \quad a_2 = 3a_0, \quad a_3 = 4a_0, \quad a_4 = 5a_0.$$

The probabilities. Since the outcome $(n+1)a_0$ is obtained precisely when the state is found in $|\phi_n\rangle$, each probability is again $|\alpha_n|^2$:

$$P(2a_0) = \frac{2}{7}, \quad P(3a_0) = \frac{3}{7}, \quad P(4a_0) = \frac{1}{7}, \quad P(5a_0) = \frac{1}{7}.$$

Collecting the results:

Value of A	$2a_0$	$3a_0$	$4a_0$	$5a_0$
Probability	$2/7$	$3/7$	$1/7$	$1/7$

Check. The probabilities again sum to one, and

$$\langle \hat{A} \rangle = \frac{2(2) + 3(3) + 1(4) + 1(5)}{7} a_0 = \frac{22}{7} a_0 \approx 3.1 a_0,$$

which lies between $2a_0$ and $5a_0$ as required.

(c) Measuring A immediately after obtaining the energy $4\mathcal{E}_0$. **(Canvas Q13)**

Step 1: identify the state after the energy measurement. The measured value $4\mathcal{E}_0$ must be one of the eigenvalues $n^2\mathcal{E}_0$, so

$$n^2\mathcal{E}_0 = 4\mathcal{E}_0 \implies n^2 = 4 \implies n = 2,$$

the negative root being discarded because the states are labelled by $n = 1, 2, 3, \dots$. No other value of n gives this energy, so the outcome identifies the eigenstate uniquely; the spectrum is non-degenerate here.

Step 2: apply the collapse postulate. Immediately after a measurement that yields the eigenvalue λ_n , the system is left in the corresponding normalised eigenstate. The superposition is therefore destroyed and the state becomes

$$|\psi_0\rangle \longrightarrow |\phi_2\rangle.$$

Step 3: measure A in that state. Acting with $\hat{A}$ on the collapsed state,

$$\hat{A}|\phi_2\rangle = (2+1)a_0|\phi_2\rangle = 3a_0|\phi_2\rangle,$$

so $|\phi_2\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $3a_0$. Applying the Born rule to this state, the expansion contains the single term $|\phi_2\rangle$ with coefficient 1, so

$$P(3a_0) = |\langle\phi_2|\phi_2\rangle|^2 = 1,$$

and every other outcome has probability zero. The measurement of A therefore returns $3a_0$ with certainty.

Check. This is consistent with part (b): $3a_0$ is one of the four values found there, and it was the outcome associated with $|\phi_2\rangle$, which is exactly the state the system has been left in. What has changed is not the list of possible values but their probabilities, which have collapsed from $(\frac{2}{7}, \frac{3}{7}, \frac{1}{7}, \frac{1}{7})$ to $(0, 1, 0, 0)$.

Final answer

- (a) The energy measurement yields $\mathcal{E}_0, 4\mathcal{E}_0, 9\mathcal{E}_0, 16\mathcal{E}_0$ with probabilities $\frac{2}{7}, \frac{3}{7}, \frac{1}{7}, \frac{1}{7}$ respectively.
- (b) The measurement of A yields $2a_0, 3a_0, 4a_0, 5a_0$ with the same probabilities $\frac{2}{7}, \frac{3}{7}, \frac{1}{7}, \frac{1}{7}$, because $\hat{A}$ and $\hat{H}$ share the eigenstates $|\phi_n\rangle$.
- (c) The energy result $4\mathcal{E}_0$ corresponds to $n = 2$ and collapses the state to $|\phi_2\rangle$. An immediate measurement of A then gives $3a_0$ with probability 1.

Physical insight

The thread running through all three parts is that $\hat{H}$ and $\hat{A}$ are *compatible* observables. Both are diagonal in the same basis $\{|\phi_n\rangle\}$, so they commute: acting with them in either order on any $|\phi_n\rangle$ gives $n^2(n+1)\mathcal{E}_0a_0|\phi_n\rangle$, hence $[\hat{H}, \hat{A}] = 0$. This is what makes part (b) almost free once part (a) is done, since no re-expansion of the state is required, and it is what makes part (c) certain rather than probabilistic.

Part (c) is the sharpest illustration. Before any measurement, neither the energy nor A has a definite value; the system is in a superposition and both are genuinely indeterminate. The energy measurement does not merely reveal a pre-existing value, it *prepares* the system in $|\phi_2\rangle$. Because that state is simultaneously an eigenstate of $\hat{A}$, the subsequent measurement of A is fully determined. Measuring one observable has removed the uncertainty in the other.

This is exactly the behaviour that fails for incompatible observables. Position and momentum, whose commutator $[\hat{x}, \hat{p}] = i\hbar$ was computed in Question 3, share no eigen-

states at all, so sharpening one necessarily blurs the other. The contrast between the two situations is the practical meaning of a commutator: it is zero when two quantities can be known together, and non-zero when they cannot. The word “immediately” in the question carries this weight, since the state $|\phi_2\rangle$ subsequently evolves in time, and a delayed measurement of an observable that did not commute with $\hat{H}$ would no longer be certain.